

Omar Ayman Bakr

Machine Learning Engineer

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SUMMARY

Mechatronics Engineering graduate specializing in Machine Learning and Computer Vision, with 500+ LeetCode problems solved. Experienced across the full ML lifecycle — data engineering, model training, evaluation, and serving behind a REST API. Currently building a retrieval-augmented document Q&A service end to end.

TECHNICAL SKILLS

- **Languages:** Python, C++, C, MATLAB, SQL
- **Machine Learning:** PyTorch, Scikit-learn, XGBoost, LightGBM, CatBoost, NumPy, Pandas, Matplotlib, Seaborn
- **Deep Learning:** CNNs (ResNet), LSTMs, Transfer Learning, Focal Loss, Hydra, TensorBoard
- **Backend & Data:** FastAPI, LangChain, MongoDB, PostgreSQL, Docker, REST APIs, async I/O
- **Tools:** Git, GitHub, Linux/Fedora, Jupyter, uv, pgAdmin4, LaTeX
- **Foundations:** Data Structures & Algorithms, Linear Algebra, Probability, Calculus

PROJECTS

Volleyball Group Activity Recognition

Computer Vision · Deep Learning

- Built eight progressively complex PyTorch baselines on the Volleyball Dataset (55 videos, 4,830 clips, 60+GB), isolating one architectural component at a time from a single-frame ResNet-50 to hierarchical player- and scene-level LSTMs with team-split pooling.
- Best model reaches **85.6% accuracy and 0.86 macro-F1**, exceeding the CVPR 2016 published result of 81.9%.

Keywords: PyTorch, ResNet-50, Hierarchical LSTM, Transfer Learning, Group Activity Recognition, Hydra, TensorBoard, LMDB, nn.DataParallel, AdamW, CosineAnnealingLR, Label Smoothing, Gradient Accumulation, Early Stopping, Macro-F1, mAP

NotebookLLM-minus — Retrieval-Augmented Document Q&A Service

LLM / RAG · Backend

- Building a NotebookLM-style backend on FastAPI and MongoDB: documents are uploaded per project, chunked with LangChain, and stored for grounded question answering; embedding and vector retrieval in active development.
- Designed a layered service (routes → controllers → models) around a single error boundary — typed exceptions carry their own HTTP status — with async Motor I/O and correlation-id logging per request.

Keywords: RAG, FastAPI, Uvicorn, MongoDB, Motor (async), LangChain, RecursiveCharacterTextSplitter, pypdf, Pydantic Settings, Docker Compose, REST API Design, Middleware, Structured Logging, uv

New York City Taxi Trip Duration Prediction

Regression · Feature Engineering

- Engineered 55 features from 10 originals by integrating five external sources (NYC DOT traffic, weather, borough geometry, holidays) and running a local OSRM routing server in Docker for real distance and travel time on 1M+ trips.
- Ridge Regression achieved **RMSE 0.397 and R^2 of 71.6%** on 224,734 held-out trips.

Keywords: Feature Engineering, OSRM Routing Engine, OpenStreetMap, Docker, Geospatial Analysis, Scikit-learn, Ridge Regression, Polynomial Features, One-Hot Encoding, Log Transformation, Outlier Removal, Pandas, NumPy, EDA

Credit Card Fraud Detection on Severely Imbalanced Data

Classification · Model Benchmarking

- Benchmarked 11 models on a 285K-transaction dataset with a $< 0.2\%$ fraud rate, implementing Focal Loss from its mathematical definition in PyTorch to attack the imbalance algorithmically rather than by resampling.
- XGBoost reached **F1 0.89 and ROC-AUC 0.98** on the minority class — a 16% F1 gain over the logistic-regression baseline.

Keywords: Imbalanced Classification, Focal Loss (custom PyTorch), XGBoost, LightGBM, CatBoost, AdaBoost, Gradient Boosting, Random Forest, KNN, Voting Ensemble, Scikit-learn, PR Curves, Average Precision, ROC-AUC, joblib

Tech Job Market Analysis with PostgreSQL

SQL · Data Analysis

- Analyzed **700K+ tech job postings** across five countries with advanced PostgreSQL — CTEs, window functions, and multi-table joins over a star schema of company, posting, and skill tables — then visualized the results in Python.
- Surfaced hiring and pay structure: data roles carry the largest senior-level salary premium, Python and SQL are near-universal requirements, and only ~8% of postings offer remote work.

Keywords: PostgreSQL, Advanced SQL, CTEs, Window Functions, Joins, Aggregations, Star Schema, pgAdmin4, Python, Pandas, Matplotlib, Seaborn, EDA, Data Visualization, Business Intelligence

EDUCATION

Bachelor of Science, Mechatronics Engineering

2019 – 2024

Helwan University, Cairo, Egypt — Graduated with Honors (Very Good, GPA 3.0/4.0)

Graduation Project: Built a [mathematical model](#) in MATLAB for a green hydrogen generator, controlling input voltage and current while predictively monitoring safety parameters such as operating temperature.